Task 3 Numerical Methods

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PROBLEM 1

5.2.30

I can use the Gershgorin Circle Theorem to define disks in which the eigenvalues λ of A must lie. For each row i, I create a disk D_i centered at the diagonal entry a_{ii} with a radius $R_i = \sum_{j \neq i} |a_{ij}|$:
D_1 is centered at 6 with radius $R_1 = 2 + 1 = 3$. Therefore, D_1 gives us $3 \leq |\lambda - 6|$, leading to $3 \leq |\lambda| \leq 9$.
D_2 is centered at -5 with radius $R_2 = 1 + 0 = 1$. Therefore, D_2 gives us $1 \leq |\lambda + 5|$, leading to $4 \leq |\lambda| \leq 6$.
D_3 is centered at 4 with radius $R_3 = 2 + 1 = 3$. Therefore, D_3 gives us $1 \leq |\lambda - 4|$, leading to $1 \leq |\lambda| \leq 7$.
By the Gershgorin Circle Theorem, all eigenvalues must lie within at least one of these disks. The intersection of these regions is $1 \leq |\lambda| \leq 9$, confirming that all eigenvalues of A must satisfy this inequality.

5.2.37

Sketch the Gershgorin disks.
For the given matrix A,

$$A = \begin{bmatrix} 0 & 2 & -1 \\ -2 & -10 & 0 \\ -1 & -1 & 4 \end{bmatrix}$$

I can apply the Gershgorin Circle Theorem to find the Gershgorin disks.

The first disk D_1 is centered at 0 with radius $R_1 = 2 + 1 = 3$. Therefore, $D_1 :| \lambda - 0| \leq 3 \Rightarrow |\lambda| \leq 3$.
The second disk D_2 is centered at -10 with radius $R_2 = 2 + 0 = 2$. Therefore, $D_2 :| \lambda + 10| \leq 2 \Rightarrow 8 \leq |\lambda| \leq 12$.
The third disk D_3 is centered at 4 with radius $R_3 = 1 + 1 = 2$. Therefore, $D_3 :| \lambda - 4| \leq 2 \Rightarrow 2 \leq |\lambda| \leq 6$.

Using the Gershgorin Circle Theorem, every eigenvalue λ of A must lie within at least one of these disks. The widest range for $|\lambda|$ is $[2, 12]$.

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Bound for Spectral Radius From Theorem 4 in the text book:,

$$\rho(A) = \inf_{\|\cdot\|} \|A\|$$

I know $\rho(A)$ is the infimum of all subordinate matrix norms of A.
Since I am provided the $\|\cdot\|_\infty$ matrix norm as defined by

$$\|A\|_\infty = \max_{1\le i\le n} \sum_{j=1}^{n} |a_{ij}|$$

I can easily find this norm for A :

$$\|A\|_\infty = \max\{3, 12, 6\} = 12$$

By Theorem 4, $\rho(A) \le \|A\|_\infty = 12$.
However, I already know from the Gershgorin disks that $\rho(A)$ must be in the range $[2, 12]$. Therefore, the bound for $\rho(A)$ is $2 \le \rho(A) \le 12$.

5.3.30

To find the least-squares solution, I will use the QR factorization method as hinted. I start by rewriting the problem in matrix form $AX = B$, where

$$A = \begin{pmatrix} 3 & 2 \\ 2 & 3 \\ 1 & 2 \end{pmatrix}, \quad X = \begin{pmatrix} x \\ y \end{pmatrix}, \quad B = \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}$$

I'll first find the QR factorization of matrix A. A can be factored into Q and R where Q is an orthogonal matrix and R is an upper triangular matrix. I have used python for these calculations.

Based on the content from page 279 , the least-squares solution X can be computed as:

$$X = R^{-1}Q^T B$$

This I have implemented in python using the np.linalg.qr method.

Upon running this code, I get:

$$X = [1.38095238, -0.66666667]$$

Therefore, the least-squares solution to the given system is x = 1.38095238 and y = -0.66666667.

5.4.1

1. Matrix A = $\begin{bmatrix} 4 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 7 \\ 0 & 0 & 0 \end{bmatrix}$

Firstly,

$$A^*A = \begin{bmatrix} 16 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 49 \end{bmatrix}$$

The eigenvalues are $\sigma_1^2 = 49, \sigma_2^2 = 16, \sigma_3^2 = 0$, so $\sigma_1 = 7, \sigma_2 = 4, \sigma_3 = 0$.

I can choose the unitary matrix Q as an identity matrix I because A^*A is already diagonal. $Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Matrix P can also be chosen as an identity matrix I. $P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Matrix D will then be: $D = \begin{bmatrix} 7 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

2. Matrix B= $\begin{bmatrix} 2 & 1 \end{bmatrix}$

$$B^*B = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$$

Eigenvalues are $\sigma_1^2 = 5, \sigma_2^2 = 0$, so $\sigma_1 = \sqrt{5}, \sigma_2 = 0$. Let's take

$$Q = \begin{bmatrix} \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$$

Since B is 1×2, P will be a 1×1 identity matrix, and D will be:

$$D = \begin{bmatrix} \sqrt{5} & 0 \end{bmatrix}$$

3. Matrix C= $\begin{bmatrix} 5 \\ -4 \end{bmatrix}$

$$C^*C = [25 + 16] = 41$$

Eigenvalue $\sigma_1^2 = 41$, so $\sigma_1 = \sqrt{41}$.

Here P will be a 2×2 identity matrix, Q will be a 1×1 identity matrix, and D will be:

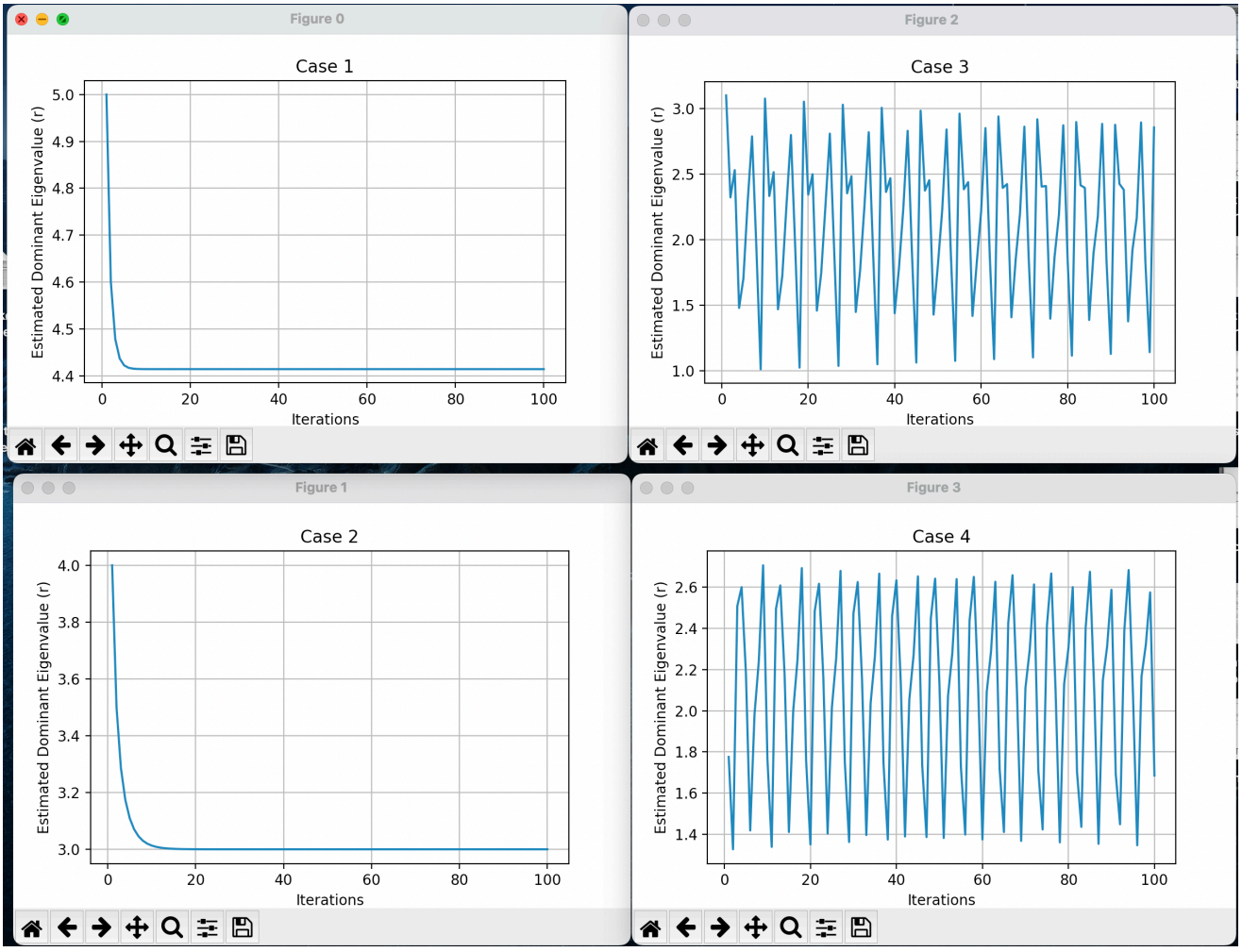
$$D = [\sqrt{41}]$$

I have now found the SVDs for all three matrices.

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PROBLEM 2

To solve this problem I have implemented the Power method for each given matrix A and initial vector x_0. I have then plotted the estimated dominant eigenvalue, r, as a function of iterations (up to 100 iterations). I have done all of this in Python. Below are the results and an explanation for each case:



1. Case 1: The plot starts high and gradually curves downwards. This indicates that the method is on a trajectory towards convergence. Given that the matrix is real and symmetric, the eigenvalue should eventually stabilize if more iterations are allowed.

2. Case 2: Similarly, this also starts from a high point and curves downwards. Even though the matrix and initial vector are complex, the behavior appears to be similar to Case 1. This suggests that the method is capable of handling complex numbers without significant difficulties in this specific case.

3. Case 3: The behavior here is notably different: the plot appears to oscillate, resembling an interference pattern. This suggests that the method is not converging. The eigenvalues of this matrix are probably complex or closely spaced, making it difficult for the power method to converge to a single dominant eigenvalue.

4. Case 4: Much like Case 3, this also shows an oscillatory behavior. The fact that both the matrix and the initial vector are complex might be causing these fluctuations, indicating a lack of convergence.